

**Quiz 02 – Chapter 02 – The System Unit: Processing and Memory [70 Marks, 60 Minutes]**

Name: \_\_\_\_\_

Roll No: \_\_\_\_\_

**Question 1: [1x12 marks] [estimated time 12 minutes]**  
**Supply the missing words to complete the statements.**

1. bit
2. Unicode
3. ALU
4. motherboard
5. Universal Serial Bus
6. PCI Express (PCIe) bus
7. 10 times
8. a system clock
9. IrDA (Infrared Data Association) and Bluetooth ports
10. multiprocessing and parallel processing.

**Question 2: [1x9 marks] [estimated time 3 minutes]**  
**Number the following terms from 1 to 9 to indicate their size from smallest to largest.**

- a) Petabyte \_\_\_\_\_6\_\_\_\_\_.
- b) Yottabyte \_\_\_\_\_9\_\_\_\_\_.
- c) Zettabyte \_\_\_\_\_8\_\_\_\_\_.
- d) Kilobyte \_\_\_\_\_2\_\_\_\_\_.
- e) Exabyte \_\_\_\_\_7\_\_\_\_\_.
- f) Terabyte \_\_\_\_\_5\_\_\_\_\_.
- g) Gigabyte \_\_\_\_\_4\_\_\_\_\_.
- h) Byte \_\_\_\_\_1\_\_\_\_\_.
- i) Megabyte \_\_\_\_\_3\_\_\_\_\_.

**Question 3: [1x5 marks] [estimated time 5 minutes]**  
**Circle T if the statement is true and F otherwise.**

1. T
2. F
3. T
4. T
5. F

**Question 4: [5x6 marks] [estimated time 25 minutes]**  
**Give Short answers to the followings****1. What is a Hot-swappable device?**

USB and FireWire devices are hot-swappable, meaning they can be plugged into their respective ports while the computer is powered up. Hot-swappable devices—along with some removable storage media, such as flash memory cards—are recognized by the computer as soon as they are connected to it and can be used right away.

**2. What is Moore's Law?**

Today's CPUs contain hundreds of millions of transistors, and the number doubles approximately every 18 months. This phenomenon is known as Moore's Law.

**3. Differentiate bandwidth and throughput?**

Bandwidth is the amount of data that can be theoretically transferred through a communications medium in a given period of time. While throughput is the amount of data that is actually transferred through a communications medium under real-life conditions.

**4. If your computer seems sluggish, list two things you could do to try to speed it up without resorting to purchasing an entirely new system.**

- i. Prevent programs from automatically running when you start up your computer.
- ii. Delete/uninstall programs you don't use.
- iii. Clean up hard disk space.
- iv. Run a disk cleanup or repair.
- v. Run a disk defragment to optimize your hard drive's efficiency.
- vi. Add more RAM to your computer.
- vii. Update your computer's software.

**5. Convert 13 in base 10 into a binary number?**

Divide by the base 2 to get the digits from the remainders:

| Division by 2 | Quotient | Remainder | Bit # |
|---------------|----------|-----------|-------|
| (13)/2        | 6        | 1         | 0     |
| (6)/2         | 3        | 0         | 1     |
| (3)/2         | 1        | 1         | 2     |
| (1)/2         | 0        | 1         | 3     |

= (1101)<sub>2</sub>

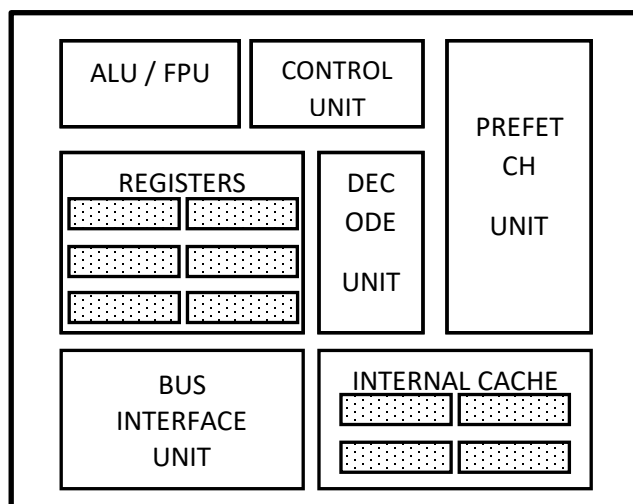
**6. Convert 1001110 in base 2 into a decimal number?**

(1001110)<sub>2</sub> =

$$(1 \times 2^7) + (0 \times 2^6) + (0 \times 2^5) + (1 \times 2^4) + (1 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) + (-16 \times 2^0)$$

= (78)<sub>10</sub>

**Question 5: [7x2 marks] [estimated time 5 minutes]**  
**Name seven components inside the CPU (at diagram)**  
**and one-line purpose of each at following lines.**



**ALU/FPU:** Performs the arithmetic and logical operations.

**CONTROL UNIT:** Is in charge of the entire process, making sure everything happens at the right time.

**PREFETCH UNIT:** Requests instructions and data from cache or RAM and makes sure they are in the proper order for processing.

**REGISTERS:** Hold the results of processing

**DECODE UNIT:** Takes instructions from the prefetch unit and translates them into a form that the control unit can understand.

**INTERNAL CACHE:** Stores data and instructions before and during processing.

**BUS INTERFACE UNIT:** The place where data and instructions enter or leave the core.